Advanced Math

Math | 27 questions with full Rule / Tip / Trap / POE / Desmos explanations

The CLC Method

Every question in this file is broken down the same way, so you build one consistent habit instead of guessing question to question:

The Rule — the underlying math principle or formula being tested. Learn the rule, not just this one answer, and you'll be able to answer every future question of this type.

The Tip — a concrete move you can make before diving into algebra, so you solve efficiently instead of the long way.

Step-by-Step Solution — every single algebra step written out in full, with nothing skipped — so you can follow exactly how the answer was reached, not just see the final number.

The Trap — the single wrong answer most students actually pick, and the specific arithmetic or logical slip that produces it.

POE (multiple-choice questions only) — a full walkthrough of the other wrong choices, showing exactly what error produces each one. Some Math questions are Student-Produced Response ('grid-in') questions with no answer choices at all — for those, there's no POE section, just the full solution.

Solving with Desmos — exact, specific instructions for using the built-in Desmos graphing calculator (available on every Digital SAT Math question) to solve or verify the question — including the exact expressions to type in, not just a general suggestion to 'graph it.'

Nonlinear functions

11 questions · Easy 2 / Medium 2 / Hard 7

Nonlinear functions — Q1 [Easy] (ID: 3cf2698e)

Question ID: 3cf2698e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The number of cells of a certain bacteria doubles every 25 minutes during its growth period when placed in a medium of milk in a controlled laboratory environment. If a sample of 830 cells of this bacteria is placed in milk in a controlled laboratory environment, which function f best represents the number of cells of the bacteria x minutes after the growth period begins, where x is less than the number of minutes in the growth period?

Answer

A. $f(x) = 830\left(\frac{1}{2}\right)^{\frac{x}{25}}$

B. $f(x) = 830(2)^{\frac{x}{25}}$

C. $f(x) = 830\left(\frac{1}{3}\right)^{\frac{x}{25}}$

D. $f(x) = 830(3)^{\frac{x}{25}}$

Correct Answer: B

The Rule: For exponential growth or decay problems, the general model is: (final amount) = (initial amount) × (growth factor)^(number of periods elapsed). When something DOUBLES every fixed time interval, the growth factor is exactly 2, and the exponent should represent 'how many complete doubling periods have occurred' — usually written as x divided by the length of one period.

The Tip: Translate the words into the model piece by piece before looking at answer choices: 'doubles' tells you the base of the exponent is 2 (not 1/2, which would be halving/decay). 'Every 25 minutes' tells you the exponent must be x/25, so that when x = 25, the exponent equals exactly 1 (one full doubling).

Why B is right: The initial amount is 830 cells. Since the population doubles (multiplies by 2) every 25 minutes, the growth factor per minute is 2^(1/25), and after x minutes the population has gone through x/25 doubling periods. This gives f(x) = 830(2)^(x/25), matching choice B.

Step-by-Step Solution:

Step 1: Identify the initial amount: 830 cells (given directly in the problem).

Step 2: Identify the growth factor: the population 'doubles,' so the base of the exponential is 2 (not 1/2 — that would represent something halving, or decaying, not growing).

Step 3: Identify the time period: doubling happens 'every 25 minutes,' so the exponent must be x/25 — this way, when x = 25 minutes have passed, the exponent is 25/25 = 1, giving exactly one full doubling.

Step 4: Assemble the full function: f(x) = (initial amount) × (growth factor)^(x / period) = 830(2)^(x/25).

Step 5: This matches choice B.

The Trap: Choice A is the most commonly picked wrong answer — it uses 1/2 as the base instead of 2, which would model the population being cut IN HALF every 25 minutes (decay) rather than doubling (growth) — a direct misreading of 'doubles' as 'halves.'

POE — Eliminate the other three:

✗ **A —** Using 1/2 as the base models decay (the population shrinking by half each period), which is the opposite of 'doubles' — this misreads growth as decay.

✗ **C —** Using 1/3 as the base doesn't correspond to anything in the problem — there's no tripling or thirding described, and it's also a decay-style base (less than 1).

✗ **D —** Using 3 as the base would model the population TRIPLING every 25 minutes, not doubling — this doesn't match the given growth rate.

Solving with Desmos:

Step 1: Open Desmos and type the function you derived: f(x)=830(2)^(x/25)

Step 2: Press Enter — Desmos will graph this exponential curve, which should start at (0, 830) and rise steeply as x increases.

Step 3: Use the table feature (click '+' then table) and enter x = 25 — the output should show f(25) = 830(2)^1 = 1660, exactly double the starting amount, confirming the doubling behavior described in the problem.

Step 4: Enter x = 50 in the table as well — you should see f(50) = 830(2)^2 = 3320, which is double 1660, confirming the population doubles again after another 25 minutes, verifying your equation and choice B.

Nonlinear functions — Q2 [Easy] (ID: fe62f031)

Question ID: fe62f031

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The function f is defined by $f(x) = 3x^5$. In the *xy*-plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ up 4 units. Which equation defines the function g?

Answer

A. $g(x) = 3x^5 + 4$

B. $g(x) = 3x^5 - 4$

C. $g(x) = 3x^{20}$

D. $g(x) = 12x^5$

Correct Answer: A

The Rule: Shifting a graph UP by a certain number of units means adding that number directly to the entire function's output — for any function y = f(x), shifting up k units gives the new function y = f(x) + k. Shifting DOWN would instead subtract k.

The Tip: Remember the direction rule: shifting UP always means ADDING a positive constant to the whole function (outside any exponents or parentheses affecting x), and shifting DOWN always means SUBTRACTING. Don't confuse this with horizontal shifts, which affect x INSIDE the function and work in the opposite, 'backwards' direction.

Why A is right: Since g is the result of shifting f up by 4 units, and f(x) = 3x^5, the new function is g(x) = f(x) + 4 = 3x^5 + 4, matching choice A.

Step-by-Step Solution:

Step 1: Write down the original function: f(x) = 3x^5.

Step 2: Recall the rule for vertical shifts: shifting a graph UP by k units means adding k to the entire function, giving g(x) = f(x) + k.

Step 3: Substitute k = 4 (since the graph shifts up 4 units) and the original function: g(x) = 3x^5 + 4.

Step 4: This matches choice A.

The Trap: Choice B is the most commonly picked wrong answer — it subtracts 4 instead of adding it, which would represent shifting the graph DOWN 4 units rather than UP, a direct sign confusion between the two vertical shift directions.

POE — Eliminate the other three:

✗ **B** — Subtracting 4 (3x^5 − 4) represents a shift DOWN 4 units, not UP — this reverses the direction described in the problem.

✗ **C** — 3x^20 changes the EXPONENT of x rather than adding a constant — this would represent a completely different type of transformation (not a simple vertical shift at all), and doesn't match 'shifting up.'

✗ **D** — 12x^5 multiplies the entire function by 4 (a vertical stretch) rather than adding 4 (a vertical shift) — stretching and shifting are different transformations.

Solving with Desmos:

Step 1: Open Desmos and type the original function: f(x)=3x^5

Step 2: Press Enter — Desmos graphs this curve.

Step 3: On the next line, type the proposed answer: g(x)=f(x)+4

Step 4: Press Enter — Desmos graphs g(x) as well, and you'll visually see it's the exact same S-shaped curve as f(x), just moved straight up by 4 units (compare where each curve crosses the y-axis: f crosses at 0, g should cross at 4).

Step 5: To double check, use the table feature for both functions at the same x-value (like x = 1): f(1) = 3, and g(1) should be exactly 3 + 4 = 7 — confirming the vertical shift is working correctly and matches choice A.

Nonlinear functions — Q3 **[Medium]** (ID: 71b74a44)

Question ID: 71b74a44

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$$y = 7{,}400(0.87)^x$$

The given equation estimates the value, in dollars, of a certain piece of equipment x years after it was purchased. According to the given equation, what was the estimated value, in dollars, of the piece of equipment at the time it was purchased?

Answer

A. 740

B. 6,438

C. 7,400

D. 8,700

Correct Answer: C

The Rule: In an exponential model of the form y = a(b)^x, the constant 'a' (the number OUTSIDE the parentheses, not raised to any power) always represents the STARTING value — the value of y when x = 0, which corresponds to 'time zero,' i.e., right at the very beginning before any time has passed.

The Tip: Whenever a question asks for the value 'at the time it was purchased' or 'initially' or 'at the start,' that's asking for the value at x = 0 — and in any exponential equation, plugging in x = 0 makes the entire exponent term equal 1 (since anything to the power of 0 is 1), leaving just the leading constant 'a' by itself.

Why C is right: At the time of purchase, x = 0 years have passed. Substituting x = 0 into y = 7,400(0.87)^x gives y = 7,400(0.87)^0 = 7,400 × 1 = 7,400, since any nonzero number raised to the power of 0 equals 1.

Step-by-Step Solution:

Step 1: Recognize that 'at the time it was purchased' means x = 0 (no years have passed yet).

Step 2: Substitute x = 0 into the given equation: y = 7,400(0.87)^0.

Step 3: Recall that any nonzero base raised to the power of 0 equals 1: (0.87)^0 = 1.

Step 4: Simplify: y = 7,400 × 1 = 7,400.

Step 5: This matches choice C.

The Trap: Choice A (740) is the most commonly picked wrong answer — it results from a decimal-place error, such as accidentally dividing 7,400 by 10, rather than correctly evaluating (0.87)^0 as exactly 1.

POE — Eliminate the other three:

✗ A — 740 is off by a factor of 10 from the correct starting value — likely from a decimal-point slip rather than correctly evaluating the zero exponent.

✗ B — 6,438 comes from actually multiplying 7,400 by 0.87 (as if x = 1, one year later), rather than using x = 0 for the very start.

✗ D — 8,700 doesn't correspond to a natural calculation from this equation — likely confused with a different number from the problem or a distractor.

Solving with Desmos:

Step 1: Open Desmos and type the given equation: y=7400(0.87)^x

Step 2: Press Enter — Desmos graphs this decreasing exponential curve.

Step 3: Use the table feature (click '+' then table) and enter x = 0 in the x-column.

Step 4: Look at the corresponding y-value in the table — it will show exactly 7400, confirming that the value at the time of purchase (x = 0) is $7,400, matching choice C.

Step 5: As an extra visual check, look at where the curve crosses the y-axis on the graph — this crossing point IS the y-intercept, which always represents the x = 0 value, and you should see it lines up with y = 7400.

Nonlinear functions — Q4 [Medium] (ID: 769b612d)

Question ID: 769b612d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

The function g is defined by $g(x) = 10^{5x-8}$. What is the value of x when $g(x)$ is equal to $10,000$?

Answer

A. $\frac{11}{5}$

B. $\frac{12}{5}$

C. 3

D. 4

Correct Answer: B

The Rule: When solving an equation where the variable is in the exponent and both sides can be written as powers of the SAME base, rewrite both sides using that common base, then set the exponents equal to each other — this converts an exponential equation into a simple linear equation you can solve directly.

The Tip: Before trying logarithms or a calculator, check whether the number on the other side of the equation (here, 10,000) can be written as a power of the SAME base already in the equation (here, base 10). If it can, this shortcut is much faster than any other method.

Why B is right: Rewrite 10,000 as a power of 10: 10,000 = 10^4. Now the equation 10^(5x−8) = 10^4 has matching bases, so the exponents must be equal: 5x − 8 = 4. Solving: 5x = 12, so x = 12/5.

Step-by-Step Solution:

Step 1: Start with the given equation: g(x) = 10^(5x−8), and we want g(x) = 10,000.

Step 2: Set up the equation: 10^(5x−8) = 10,000.

Step 3: Rewrite 10,000 as a power of 10: 10,000 = 10 × 10 × 10 × 10 = 10^4.

Step 4: Now both sides have the same base (10): 10^(5x−8) = 10^4.

Step 5: Since the bases match, set the exponents equal to each other: 5x − 8 = 4.

Step 6: Add 8 to both sides: 5x = 12.

Step 7: Divide both sides by 5: x = 12/5.

Step 8: This matches choice B.

The Trap: Choice A (11/5) is the most commonly picked wrong answer — it typically results from an arithmetic slip while solving 5x − 8 = 4, such as adding 8 incorrectly or misreading the exponent 4 as a different number during the final steps.

POE — Eliminate the other three:

✗ A — 11/5 doesn't satisfy 5x − 8 = 4 — plugging x = 11/5 back in gives 5(11/5) − 8 = 11 − 8 = 3, not 4, so this doesn't actually solve the equation correctly.

✗ **C —** 3 would come from a completely different (incorrect) exponent equation, not from correctly setting 5x − 8 = 4.

✗ **D —** 4 is the value of the EXPONENT (4) itself, not the value of x — a student who correctly identifies 10,000 = 10^4 but then stops there, forgetting to actually solve 5x − 8 = 4 for x, might mistakenly select 4.

Solving with Desmos:

Step 1: Open Desmos and type the original function: g(x)=10^(5x-8)

Step 2: Press Enter — Desmos graphs this steep exponential curve.

Step 3: To find where g(x) = 10,000, type a second equation on the next line: y=10000 — this creates a flat horizontal line at height 10,000.

Step 4: Look for where the two graphs (the curve and the horizontal line) intersect — click directly on that intersection point, and Desmos will display its exact coordinates.

Step 5: The x-coordinate of that intersection point will show 2.4 (which is the decimal form of 12/5), confirming x = 12/5 and matching choice B.

Nonlinear functions — Q5 [Hard] (ID: f01108a8)

Question ID: f01108a8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$y = 9\left(\frac{a}{7}\right)^{x+c} - b$$

How many times does the graph of the given equation in the *xy*-plane cross the *x*-axis, where a, b, and c are positive constants such that $a > 7$ and $b > c$?

Answer

A. Zero

B. One

C. Two

D. Three

Correct Answer: B

The Rule: For an exponential function of the form y = A(base)^(x+c) − b (where the base is greater than 1, making it an increasing exponential), the graph has a horizontal asymptote at y = −b and is always strictly increasing. A strictly increasing curve with a single horizontal asymptote crosses the x-axis EXACTLY ONCE if the asymptote is below the x-axis (negative), and NEVER crosses if the asymptote is above or on the x-axis.

The Tip: For these 'how many times does the graph cross the x-axis' questions with exponential functions, you don't need to solve any equation — just figure out the sign of the horizontal asymptote (the value the function approaches but never reaches) and whether the function is increasing or decreasing. This alone tells you the crossing count.

Why B is right: Since a > 7, the base a/7 is greater than 1, meaning the exponential term (a/7)^(x+c) is always positive and grows as x increases — this makes the whole function strictly increasing. As x approaches negative infinity, the exponential term approaches 0, so y approaches −b. Since b is a positive constant, −b is negative, meaning the graph's horizontal asymptote sits below the x-axis. A strictly increasing curve that starts near a negative asymptote and increases without bound must cross the x-axis exactly once.

Step-by-Step Solution:

Step 1: Identify the base of the exponential term: a/7. Since the problem states a > 7, this means a/7 > 1, so the exponential term is a GROWING (increasing) exponential, not a decaying one.

Step 2: Determine the horizontal asymptote: as x → −∞, (a/7)^(x+c) → 0, so y → 0 − b = −b.

Step 3: Since b is stated to be a positive constant, −b is a negative number — meaning the graph's horizontal asymptote is below the x-axis (in negative y territory).

Step 4: Since the function is strictly increasing (from Step 1) and approaches a NEGATIVE value as x → −∞ but grows without bound as x → +∞, the graph must cross from below the x-axis to above it — and because it's strictly increasing (never doubles back), this crossing happens exactly once.

Step 5: The condition b > c given in the problem is extra information not needed to answer this particular question — it doesn't change the crossing count, only potentially the crossing's exact location.

Step 6: This matches choice B: One.

The Trap: Choice A (Zero) is the most commonly picked wrong answer — students sometimes assume that because the function has a vertical shift (−b), it might never reach zero, without realizing that since the function grows without bound in the positive direction, it's guaranteed to eventually cross the x-axis exactly once.

POE — Eliminate the other three:

✗ A — The function grows without bound as x increases (since the base a/7 > 1), so it will always eventually rise above the x-axis, guaranteeing at least one crossing — 'zero crossings' isn't possible here.

✗ C — An exponential function of this form is always STRICTLY increasing (monotonic) when its base is greater than 1 — it can never curve back down to cross the x-axis a second time.

✗ D — Like choice C, three crossings would require the function to change direction multiple times, but a simple exponential function like this one only ever increases or only ever decreases — it can never reverse direction to cross more than once.

Solving with Desmos:

Step 1: Since this problem involves unknown constants a, b, and c, pick simple test values that satisfy the given conditions (a > 7, b > c > 0) to make the function concrete — let's use a = 14 (so a/7 = 2), c = 1, and b = 5. Step 2: Open Desmos and type: y=9(14/7)^(x+1)-5, which simplifies to y=9(2)^(x+1)-5

Step 3: Press Enter — Desmos graphs this curve.

Step 4: Look at how many times the curve crosses the horizontal x-axis — you should see it cross exactly ONCE, confirming choice B.

Step 5: Try a few different valid values for a, b, and c (always keeping a > 7 and b > c > 0) and re-graph — you'll find the graph always crosses the x-axis exactly once, no matter which valid values you choose, confirming this is a general property of the function's structure, not just a coincidence of your specific numbers.

Nonlinear functions — Q6 [Hard] (ID: 58b4c6f3)

Question ID: 58b4c6f3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the *xy*-plane passes through the points $(0, 10)$ and $(2, 13)$. What is the value of ab?

Answer

A. 13

B. 18

C. 20

D. 26

Correct Answer: B

The Rule: When a function has unknown constants and you're given two specific points the graph passes through, substitute each point's x and y values into the function to create a system of two equations — then solve that system for the unknown constants.

The Tip: Always substitute the point with $x = 0$ first if one is given — since anything raised to the power of 0 equals 1, this usually eliminates the exponential term entirely and gives you a much simpler equation to start solving with.

Why B is right: Substituting (0, 10): $h(0) = a^0 + b = 1 + b = 10$, so $b = 9$. Substituting (2, 13): $h(2) = a^2 + b = a^2 + 9 = 13$, so $a^2 = 4$, giving $a = 2$ (taking the positive root, since a is stated to be a positive constant). Therefore $ab = 2 \times 9 = 18$.

Step-by-Step Solution:

Step 1: Write down the general function: $h(x) = a^x + b$.

Step 2: Substitute the first point (0, 10): $h(0) = a^0 + b$. Since any nonzero number to the power of 0 equals 1, this simplifies to $1 + b = 10$.

Step 3: Solve for b: $b = 10 - 1 = 9$.

Step 4: Substitute the second point (2, 13): $h(2) = a^2 + b = a^2 + 9 = 13$.

Step 5: Solve for a^2: $a^2 = 13 - 9 = 4$.

Step 6: Take the square root of both sides: $a = 2$ (using the positive root, since the problem states a is a positive constant).

Step 7: Calculate the requested value: $ab = 2 \times 9 = 18$.

Step 8: This matches choice B.

The Trap: Choice A (13) is the most commonly picked wrong answer — it comes from confusing the requested product ab with one of the given y-coordinates (13) from the original points, rather than actually solving for and multiplying the two constants.

POE — Eliminate the other three:

✗ A — 13 is one of the given y-coordinates from the original points (0, 10) and (2, 13) — it's not the product ab, which requires solving for both constants first.

✗ C — 20 doesn't correspond to a natural calculation error from this problem — likely a distractor value.

✗ D — 26 might result from adding a and b incorrectly derived values instead of multiplying them, or from a sign/arithmetic slip while solving for a^2.

Solving with Desmos:

Step 1: Open Desmos. Step 2: Type the two given points directly to visualize them: (0,10) and (2,13) — Desmos will plot these as dots.

Step 3: Type the general function with sliders: h(x)=a^x+b — Desmos will offer to create sliders for a and b automatically (click 'all' when prompted, or add sliders manually).

Step 4: Adjust the sliders for a and b until the curve h(x) passes exactly through both plotted points — you should find this happens when a = 2 and b = 9.

Step 5: With the sliders confirmed at a = 2 and b = 9, type a new line: 2*9 (or reference the slider values directly) — Desmos will show 18, confirming ab = 18 and matching choice B.

Nonlinear functions — Q7 **[Hard]** (ID: 7bbfbe70)

Question ID: 7bbfbe70

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A lab analyst observes a sample of a substance. An exponential model estimates that the mass, in grams, of the sample decreases by 24% every 22.96 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the lab analyst began observing the sample?

Answer

A. $M = 100(0.24)^{t+22.96}$

B. $M = 100(0.24)^{\frac{t}{22.96}}$

C. $M = 100(0.76)^{t+22.96}$

D. $M = 100(0.76)^{\frac{t}{22.96}}$

Correct Answer: D

The Rule: When a quantity 'decreases by p%' repeatedly, the correct decay factor to use in an exponential model is (1 − p/100), NOT p/100 itself — because after each period, the quantity retains (100 − p)% of its previous value, not just p%.

The Tip: Never use the percentage given in a 'decreases by p%' problem directly as the base of your exponential — always convert it first by subtracting from 100% (or from 1, in decimal form). A common trap answer will use the raw percentage as if it were the retained amount instead of the LOST amount.

Why D is right: Since the mass decreases by 24% every 22.96 minutes, it RETAINS 100% − 24% = 76% of its mass each period, which as a decimal is 0.76. The exponent must be t/22.96 so that after t = 22.96 minutes, exactly one full decay period has occurred. This gives M = 100(0.76)^(t/22.96), matching choice D.

Step-by-Step Solution:

Step 1: Identify the percentage decrease: the mass decreases by 24% every 22.96 minutes.

Step 2: Convert this to a retention rate (the fraction of mass that REMAINS after each period): 100% − 24% = 76%, or as a decimal, 0.76.

Step 3: Identify the time period for the exponent: since the decay happens 'every 22.96 minutes,' the exponent should be t/22.96, so that after exactly 22.96 minutes, the exponent equals 1 (one full decay period).

Step 4: Assemble the model: M = (initial mass) × (retention rate)^(t / period) = 100(0.76)^(t/22.96).

Step 5: This matches choice D.

The Trap: Choice B is the most commonly picked wrong answer — it correctly uses t/22.96 as the exponent, but incorrectly uses 0.24 (the percentage LOST) as the base instead of 0.76 (the percentage RETAINED), which would model the mass shrinking to 24% of its PREVIOUS value every period — an extremely fast decay, not a 24% decrease.

POE — Eliminate the other three:

✗ A — This uses 0.24 as the base (should be 0.76) AND adds 22.96 directly to the exponent instead of dividing by it — both the base and the exponent structure are incorrect.

✗ B — This uses 0.24 as the base — but 0.24 represents the percentage LOST, not the percentage RETAINED; using it directly would model the mass dropping to just 24% of its value every period, a much faster decay than described.

✗ C — This uses the correct base (0.76) but the wrong exponent structure — adding 22.96 directly to t instead of dividing t by 22.96 doesn't correctly represent 'one full decay period every 22.96 minutes.'

Solving with Desmos:

Step 1: Open Desmos and type the equation you derived: M=100(0.76)^(t/22.96)

Step 2: Press Enter — Desmos graphs this decreasing exponential curve.

Step 3: Use the table feature and enter t = 22.96 — the output should show M = 100(0.76)^1 = 76, confirming the mass has dropped to 76% of its original value (a 24% decrease) after exactly one period, matching the problem's description.

Step 4: Enter t = 45.92 (double the period) in the table as well — you should see M = 100(0.76)^2 = 57.76, confirming the mass decreases by another 24% of its NEW value during the second period, verifying the exponential decay model and choice D.

Nonlinear functions — Q8 [Hard] (ID: a1e65979)

Question ID: a1e65979

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is a quadratic function. In the xy-plane, the graph of $y = f(x)$ has a vertex at $(1, 9)$ and passes through the points $(2, 33)$ and $(-1, 105)$. What is the value of $f(-2) - f(0)$?

Answer

A. 81

B. 129

C. 192

D. 225

Correct Answer: C

The Rule: For a quadratic function given in vertex form y = a(x − h)^2 + k, where (h, k) is the vertex, use ONE additional known point on the graph to solve for the leading coefficient 'a' — then you have the complete equation and can evaluate the function at any x-value needed.

The Tip: When a quadratic's vertex is given directly, always start by writing the function in VERTEX FORM (using the vertex coordinates), not standard form — vertex form requires solving for only ONE unknown (the coefficient a) using a single additional point, which is much more efficient than solving a system with three unknowns.

Why C is right: With vertex (1, 9), write f(x) = a(x − 1)^2 + 9. Use the point (2, 33): 33 = a(2−1)^2 + 9 = a(1) + 9, so a = 24. This gives f(x) = 24(x−1)^2 + 9. Now calculate f(−2) = 24(−2−1)^2 + 9 = 24(9) + 9 = 216 + 9 = 225, and f(0) = 24(0−1)^2 + 9 = 24(1) + 9 = 33. Finally, f(−2) − f(0) = 225 − 33 = 192.

Step-by-Step Solution:

Step 1: Write the quadratic in vertex form using the given vertex (1, 9): f(x) = a(x − 1)^2 + 9.

Step 2: Use the given point (2, 33) to solve for a: substitute x = 2 and f(x) = 33: 33 = a(2 − 1)^2 + 9, which simplifies to 33 = a(1) + 9, so 33 = a + 9.

Step 3: Solve for a: a = 33 − 9 = 24.

Step 4: Write the complete function: f(x) = 24(x − 1)^2 + 9.

Step 5: (Optional check using the third point (−1, 105)): f(−1) = 24(−1−1)^2 + 9 = 24(4) + 9 = 96 + 9 = 105. This matches the given point, confirming the equation is correct.

Step 6: Calculate f(−2): f(−2) = 24(−2−1)^2 + 9 = 24(−3)^2 + 9 = 24(9) + 9 = 216 + 9 = 225.

Step 7: Calculate f(0): f(0) = 24(0−1)^2 + 9 = 24(−1)^2 + 9 = 24(1) + 9 = 24 + 9 = 33.

Step 8: Calculate the requested difference: f(−2) − f(0) = 225 − 33 = 192.

Step 9: This matches choice C.

The Trap: Choice A (81) is the most commonly picked wrong answer — it results from subtracting the x-values instead of the function outputs, or from a similar mix-up between the INPUT difference and the actual OUTPUT (y-value) difference the question is asking for.

POE — Eliminate the other three:

✗ **A —** 81 doesn't come from correctly evaluating f(−2) and f(0) and subtracting — it likely results from a calculation error, such as using the wrong vertex offset value.

✗ **B —** 129 similarly doesn't align with the correct values of f(−2) = 225 and f(0) = 33 — likely from an arithmetic slip in one of the intermediate steps.

✗ **D —** 225 is the value of f(−2) ALONE, not the full difference f(−2) − f(0) — a student who correctly calculates f(−2) but forgets to subtract f(0) would mistakenly select this.

Solving with Desmos:

Step 1: Open Desmos and type the function in vertex form using the given vertex: f(x)=a(x-1)^2+9 — Desmos will offer to create a slider for 'a' (click to add it).

Step 2: Also type the two given points to plot them: (2,33) and (-1,105)

Step 3: Adjust the slider for 'a' until the curve f(x) passes through both plotted points — you should find this happens at a = 24.

Step 4: Once confirmed, type f(-2) directly into Desmos — it will evaluate to 225.

Step 5: Type f(0) on the next line — it will evaluate to 33.

Step 6: Type f(-2)-f(0) on a final line — Desmos will compute this as 192, confirming choice C.

Nonlinear functions — Q9 [Hard] (ID: 5acbdc30)

Question ID: 5acbdc30

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is defined by $f(x) = 56(0.19)^x$. For any positive integer n, the value of $f(n)$ is $p\%$ less than the value of $f(n-1)$. What is the value of p?

Answer

A. 19

B. 44

C. 56

D. 81

Correct Answer: D

The Rule: In an exponential function y = a(b)^x where the base b is a decimal between 0 and 1, the percent DECREASE from one integer x-value to the next is always (1 − b) × 100%, regardless of which specific integer values of x you compare — this percent change is constant for all exponential functions of this form.

The Tip: For 'percent change from f(n−1) to f(n)' questions with an exponential function, you never need to plug in actual numbers for n — the percent change only depends on the base of the exponential (the number being raised to the x power), which you can read directly from the function.

Why D is right: The function f(x) = 56(0.19)^x has a base of 0.19, meaning each time x increases by 1, the output is multiplied by 0.19 — so the new value retains only 19% of the previous value. This means the value DECREASES by 100% − 19% = 81% each time, so p = 81.

Step-by-Step Solution:

Step 1: Identify the base of the exponential function: 0.19 (the number being raised to the power of x).

Step 2: Understand what multiplying by 0.19 means: each time x increases by 1, the new value is only 19% (0.19) of the previous value — meaning 19% of the value is RETAINED.

Step 3: Calculate the percent that is LOST (the percent decrease): 100% − 19% = 81%.

Step 4: Since f(n) is p% less than f(n−1), and we just calculated this percent decrease is 81%, we have p = 81.

Step 5: This matches choice D.

(Algebraic verification: f(n)/f(n−1) = 56(0.19)^n / 56(0.19)^(n−1) = 0.19^1 = 0.19. This ratio means f(n) = 0.19 × f(n−1), so f(n) is 19% OF f(n−1), meaning it decreased by 100% − 19% = 81% from f(n−1).)

The Trap: Choice A (19) is the most commonly picked wrong answer — it directly uses the base of the exponential (0.19, or 19%) as the answer, without realizing that 19% represents the amount RETAINED after each step, not the amount DECREASED — these are two different, complementary percentages.

POE — Eliminate the other three:

✗ A — 19 is the percent of the value that REMAINS (is retained) after each step — but the question asks for the percent DECREASE, which is the complement: 100 − 19 = 81, not 19 itself.

✗ B — 44 doesn't correspond to any natural calculation from the base 0.19 — likely a distractor value.

✗ C — 56 is simply the leading coefficient (the initial value) of the function — it has nothing to do with the percent change between consecutive terms, which depends only on the base.

Solving with Desmos:

Step 1: Open Desmos and type the function: f(x)=56(0.19)^x

Step 2: Press Enter — Desmos graphs this rapidly decreasing exponential curve.

Step 3: Use the table feature and enter x = 0 and x = 1 to see two consecutive values — you'll see f(0) = 56 and f(1) = 10.64.

Step 4: Calculate the percent decrease directly: type (56-10.64)/56*100 into Desmos — it will evaluate to 81, confirming p = 81 and matching choice D.

Step 5: To confirm this percent decrease is CONSTANT (the same between any two consecutive integers, not just 0 and 1), try the same calculation with x = 5 and x = 6 in the table — you'll find the percent decrease between f(5) and f(6) is also exactly 81%, confirming this is a general property of the exponential base, not specific to any particular starting point.

Nonlinear functions — Q10 [Hard] [Grid-In] (ID: c95686e9)

Question ID: c95686e9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$h(t) = -16t^2 + b$$

The function h estimates a ball's height, in feet, above a floor t seconds after the ball is released, where b is a constant. The function estimates that the ball is 19.36 feet above the floor when it is released at $t = 0$. How many seconds after being released does the function estimate the ball will reach the floor?

Correct Answer: 1.1 (Student-Produced Response — no answer choices given)

The Rule: When a real-world quadratic model gives you height above the ground over time, 'reaching the floor' means the height equals 0 — so set the function equal to 0 and solve for the positive value of t (since negative time doesn't make sense in this context).

The Tip: First use the given starting condition (the height at t = 0) to solve for any unknown constant in the function, THEN set the complete function equal to 0 and solve for t — don't try to do both steps at once.

Why this is the answer: Since the ball is 19.36 feet above the floor at t = 0, substitute into h(t) = −16t^2 + b: h(0) = −16(0)^2 + b = b = 19.36, so b = 19.36. Now set the height to 0 to find when the ball reaches the floor: 0 = −16t^2 + 19.36. Solving: 16t^2 = 19.36, so t^2 = 19.36/16 = 1.21, giving t = √1.21 = 1.1 (taking the positive square root, since time must be positive).

Step-by-Step Solution:

Step 1: Use the given initial condition to find the constant b: at t = 0, the height is 19.36 feet, so substitute into h(t) = −16t^2 + b: h(0) = −16(0)^2 + b = b. Since h(0) = 19.36, this means b = 19.36.

Step 2: Write the complete function: h(t) = −16t^2 + 19.36.

Step 3: The ball reaches the floor when its height equals 0, so set h(t) = 0: −16t^2 + 19.36 = 0.

Step 4: Add 16t^2 to both sides: 19.36 = 16t^2.

Step 5: Divide both sides by 16: t^2 = 19.36/16 = 1.21.

Step 6: Take the square root of both sides: t = √1.21 = 1.1 (using the positive root, since time elapsed must be a positive value).

Step 7: This is a grid-in question — enter 1.1 as your final answer.

The Trap: A common error on this problem is forgetting to take the SQUARE ROOT at the final step, mistakenly submitting t^2 = 1.21 as the final answer instead of continuing to solve for t = 1.1, or alternatively including the negative root (−1.1), which doesn't make sense for a time value.

Solving with Desmos:

Step 1: Open Desmos and type the equation with the constant already substituted: y=-16x^2+19.36 (using x in place of t for graphing purposes).

Step 2: Press Enter — Desmos graphs this downward-opening parabola.

Step 3: Look for where the curve crosses the x-axis (where height = 0) on the POSITIVE side of the x-axis (since negative time doesn't make sense here) — click on that point.

Step 4: Desmos will display the exact coordinates of that crossing point, showing x = 1.1 (with y = 0), confirming the ball reaches the floor at t = 1.1 seconds.

Step 5: Enter 1.1 into the grid-in answer box.

Nonlinear functions — Q11 [Hard] (ID: 7bf77c19)

Question ID: 7bf77c19

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The functions p and r are defined by the equations shown, where a and b are integer constants such that $a < b$ and $b < 0$. If $y = p(x)$ and $y = r(x)$ are graphed in the *xy*-plane, which of the following equations displays, as a constant or coefficient, the *y*-coordinate of the *y*-intercept of the graph of the corresponding function?

I. $p(x) = a(4.1)^x + b$

II. $r(x) = a(4.1)^{x+b}$

Answer

A. I only

B. II only

C. I and II

D. Neither I nor II

Correct Answer: D

The Rule: The y-intercept of any function is its value when x = 0. For it to be directly 'displayed as a constant or coefficient' in an equation (without needing any further calculation), the equation must be written so that setting x = 0 leaves ONLY one of the already-visible numbers by itself — not a combination or product of multiple constants.

The Tip: For each candidate equation, mentally plug in x = 0 and simplify as far as possible. If what's left over is just ONE of the letters/constants already sitting alone in the original equation (not added to or multiplied by anything else), that equation displays its y-intercept directly. If you still need to compute a sum or product to get the final value, it does NOT count as being 'displayed.'

Why D is right: For equation I, $p(0) = a(4.1)^0 + b = a(1) + b = a + b$ — this requires ADDING two constants together, so the y-intercept isn't a single constant or coefficient already sitting in the equation by itself. For equation II, $r(0) = a(4.1)^{(0+b)} = a(4.1)^b$ — this requires MULTIPLYING a by a power of 4.1, again not a single displayed constant. Since neither equation shows its y-intercept as a single already-visible constant or coefficient, the answer is 'Neither I nor II.'

Step-by-Step Solution:

Step 1: For equation I, $p(x) = a(4.1)^x + b$, substitute $x = 0$: $p(0) = a(4.1)^0 + b$.

Step 2: Simplify using the rule that anything to the power of 0 equals 1: $p(0) = a(1) + b = a + b$.

Step 3: Notice that $a + b$ is a SUM of two separate constants — it's not simply 'a' or simply 'b' displayed alone; you'd need to actually calculate this sum to know the y-intercept's value. So equation I does NOT directly display its y-intercept as a single constant or coefficient.

Step 4: For equation II, $r(x) = a(4.1)^{(x+b)}$, substitute $x = 0$: $r(0) = a(4.1)^{(0+b)} = a(4.1)^b$.

Step 5: This is a PRODUCT of a and a power of 4.1 raised to b — again, not simply 'a' or simply 'b' by itself; you'd need to calculate this product to find the actual y-intercept value. So equation II also does NOT directly display its y-intercept.

Step 6: Since neither equation shows its y-intercept as an already-visible single constant or coefficient, the correct answer is D: Neither I nor II.

The Trap: Choice C (I and II) is the most commonly picked wrong answer — students sometimes assume that because both equations CONTAIN the letters a and b, one of those letters must represent the y-intercept directly, without carefully checking whether evaluating at $x = 0$ actually isolates just one constant or requires further calculation (addition or multiplication).

POE — Eliminate the other three:

✗ A — Equation I's y-intercept is $a + b$ (a sum), not simply 'a' or 'b' displayed alone — you still need to add two numbers together to find it, so it's not directly 'displayed.'

✗ B — Equation II's y-intercept is $a(4.1)^b$ (a product involving an exponent), not simply 'a' or 'b' displayed alone — significant calculation is still needed to find the actual value.

✗ C — Since NEITHER equation isolates its y-intercept as a single already-visible constant (both require additional calculation — addition for I, multiplication with an exponent for II), 'I and II' incorrectly claims both work when neither actually does.

Solving with Desmos:

Step 1: Since this problem involves abstract reasoning about GENERAL constants a and b rather than specific numbers, pick simple test values to verify your reasoning — let's use $a = 2$ and $b = -3$ (remembering $b < 0$ as given). Step 2: Open Desmos and type equation I with these values: y=2(4.1)^x+(-3)

Step 3: Use the table feature and enter $x = 0$ — the output will show $y = 2(1) - 3 = -1$. Notice this value (-1) doesn't match either a (2) or b (−3) directly — it required computing $a + b$, confirming equation I doesn't display its y-intercept as a simple constant.

Step 4: Now type equation II with the same values: y=2(4.1)^(x+(-3))

Step 5: Enter $x = 0$ in the table — the output will show some computed decimal value (from $2 \times 4.1^{(-3)}$), which again doesn't match a or b directly, confirming equation II also requires calculation, not direct display — verifying choice D.

Nonlinear equations in one variable and systems of equations in two variables

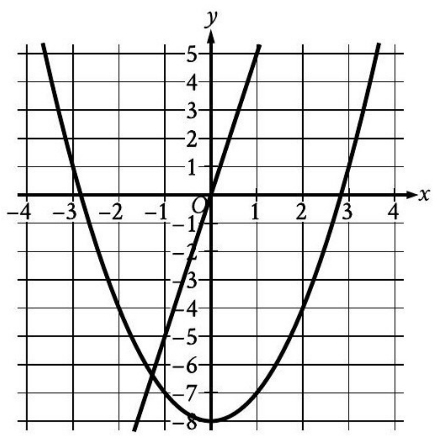
9 questions · Easy 1 / Medium 2 / Hard 6

Nonlinear equations in one variable and systems of equations in two variables — Q1 **[Easy]** (ID: f123e039)

Question ID: f123e039

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question



The graph of a system of a linear and a quadratic equation is shown. Which system of equations is represented by the graph?

Answer

A. $y = -5x$

$y = x^2 + 8$

B. $y = -5x$

$y = x^2 - 8$

C. $y = 5x$

$y = x^2 + 8$

D. $y = 5x$

$y = x^2 - 8$

Correct Answer: B

The Rule: To match a graphed parabola or line to its equation, first read the KEY FEATURES directly off the graph — for a parabola, its vertex (lowest or highest point); for a line, its y-intercept and slope — then compare these features to each answer choice.

The Tip: Start with the parabola's vertex — it's usually the easiest feature to spot exactly on a grid. The vertex's y-coordinate directly tells you the constant term in the parabola's equation when written as y = x^2 + (constant), instantly narrowing your choices.

Why B is right: The parabola's vertex sits at (0, −8), meaning its equation is y = x^2 − 8 (matching the pattern y = x^2 + k, where k = −8 is the vertical shift). The line passes through the origin and through approximately (−1, 5), giving a slope of (5−0)/(−1−0) = −5, so its equation is y = −5x. Together, these match choice B.

Step-by-Step Solution:

Step 1: Identify the parabola's vertex directly from the graph: it appears to be at (0, −8), the lowest point of the U-shaped curve.

Step 2: Since a basic parabola y = x^2 shifted down by 8 units has equation y = x^2 − 8, and the vertex confirms this shift, the parabola's equation is y = x^2 − 8.

Step 3: Identify a second clear point on the line: it appears to pass through the origin (0, 0) and through approximately (−1, 5).

Step 4: Calculate the line's slope using these two points: slope = (5 − 0) / (−1 − 0) = 5 / (−1) = −5.

Step 5: Since the line passes through the origin with slope −5, its equation is y = −5x.

Step 6: Compare both equations to the answer choices: y = −5x and y = x^2 − 8 matches choice B exactly.

The Trap: Choice A is the most commonly picked wrong answer — it uses the correct line equation but the WRONG sign on the parabola's vertical shift (+8 instead of −8), which would place the parabola's vertex ABOVE the x-axis instead of below it, not matching the graph shown.

POE — Eliminate the other three:

✗ A — This uses y = x^2 + 8, which would place the parabola's vertex at (0, 8) — well above the x-axis — but the graph clearly shows the vertex below the x-axis, around y = −8.

✗ C — This uses y = 5x for the line, which has a POSITIVE slope — but the line in the graph is clearly decreasing (going down from left to right), meaning it needs a NEGATIVE slope.

✗ D — This combines the wrong-sign line (positive slope 5x instead of −5x) with the correct parabola — but the line's direction doesn't match the graph, which shows a decreasing line.

Solving with Desmos:

Step 1: Open Desmos and type the system you determined: y=-5x and on a second line y=x^2-8

Step 2: Press Enter after each — Desmos will graph both.

Step 3: Compare this graph directly to the picture given in the question — check that the parabola's lowest point matches (0, −8) and that the line passes through the origin, sloping steeply downward to the right.

Step 4: If everything matches the original picture, you've confirmed choice B is correct. If you're ever unsure between two candidate systems, graph both pairs in Desmos (they'll appear in different colors) and see which pair visually overlaps the original picture most closely.

Nonlinear equations in one variable and systems of equations in two variables — Q2 [Medium] [Grid-In] (ID: 6bd40794)

Question ID: 6bd40794

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$$(x-16)(x-10)(x+7)(x+17)=0$$

What is a positive solution to the given equation?

Correct Answer: 16 (or 10) (Student-Produced Response — no answer choices given)

The Rule: For an equation already written as a product of factors set equal to zero, the Zero Product Property says the equation is true whenever ANY ONE of the individual factors equals zero — so you can find every solution by setting each factor equal to zero separately and solving.

The Tip: Never expand or multiply out a fully-factored equation like this one — that would create unnecessary work. Instead, immediately set each individual factor equal to zero, one at a time, to instantly find each solution.

Why this is the answer: Setting each factor equal to zero: $(x - 16) = 0$ gives $x = 16$; $(x - 10) = 0$ gives $x = 10$; $(x + 7) = 0$ gives $x = -7$; $(x + 17) = 0$ gives $x = -17$. The positive solutions are 16 and 10 — either one is an acceptable grid-in answer.

Step-by-Step Solution:

Step 1: Recognize that the equation is already fully factored: $(x - 16)(x - 10)(x + 7)(x + 17) = 0$.

Step 2: Apply the Zero Product Property: since the product of these four factors equals 0, at least one of the factors itself must equal 0.

Step 3: Set each factor equal to 0 individually and solve: $x - 16 = 0$ gives $x = 16$; $x - 10 = 0$ gives $x = 10$; $x + 7 = 0$ gives $x = -7$; $x + 17 = 0$ gives $x = -17$.

Step 4: The complete solution set is $\{16, 10, -7, -17\}$. The question asks for a POSITIVE solution — both 16 and 10 qualify.

Step 5: This is a grid-in question — enter either 16 or 10 as your final answer; both are accepted as correct.

The Trap: A common error on this type of problem is accidentally flipping the sign when setting a factor to zero — for example, incorrectly solving $(x - 16) = 0$ as $x = -16$ instead of $x = 16$, which would produce a negative value instead of the correct positive solution.

Solving with Desmos:

Step 1: Open Desmos and type the equation exactly as given: (x-16)(x-10)(x+7)(x+17)=0

Step 2: Press Enter — since this is a one-variable equation, Desmos will find and plot ALL solutions as labeled points along the x-axis.

Step 3: You should see four labeled points: $x = -17$, $x = -7$, $x = 10$, and $x = 16$.

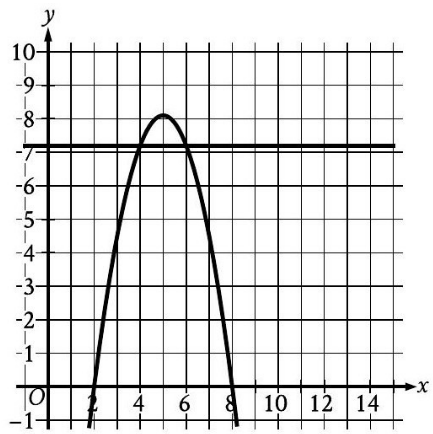
Step 4: Since the question asks for a POSITIVE solution, choose either $x = 10$ or $x = 16$ and enter it into the grid-in box — both are correct answers.

Nonlinear equations in one variable and systems of equations in two variables — Q3 [Medium] (ID: 5f10c095)

Question ID: 5f10c095

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question



The graph of a system of a linear equation and a quadratic equation is shown. A solution to the system is (x, y). What is a possible value of x?

Answer

A. 4

B. 5

C. 7.2

D. 8

Correct Answer: A

The Rule: When a graph shows the intersection of a line and a curve, the x-coordinate of ANY intersection point is automatically a valid solution to the system those two graphs represent — you can read this directly off the graph without doing any algebra, as long as you can identify the intersection points clearly.

The Tip: When answer choices give you SPECIFIC numbers to check against a graph, don't try to find the exact equations of the curves first — just look directly at the graph for where the two curves cross, and check which of the given x-values lines up with a crossing point.

Why A is right: The horizontal line y = 7 intersects the parabola at two points, occurring at approximately x = 4 and x = 6 (visually symmetric around the parabola's vertex, which appears to be at x = 5). Since x = 4 is one of these intersection points, it's a valid solution to the system, matching choice A.

Step-by-Step Solution:

Step 1: Identify the horizontal line's equation from the graph: it appears to be a flat line at y = 7.

Step 2: Look for where this horizontal line crosses the parabola — these crossing points are the solutions to the system.

Step 3: Reading the graph carefully, the line and parabola appear to intersect at approximately x = 4 and x = 6 (the parabola's vertex is at approximately (5, 8), and the two intersection points are symmetric around this vertex).

Step 4: Compare these x-values to the answer choices: x = 4 is listed as choice A, confirming it as a valid solution.

The Trap: Choice C (7.2) is the most commonly picked wrong answer — it confuses the y-coordinate of the horizontal line (y = 7) with a required x-coordinate, creating a plausible-looking decimal that doesn't actually correspond to any point where the graphs intersect.

POE — Eliminate the other three:

✗ **B** — $x = 5$ corresponds to the parabola's VERTEX (its highest point, around $y = 8$), not to where it crosses the horizontal line $y = 7$ — at $x = 5$, the parabola's y-value doesn't equal 7.

✗ **C** — 7.2 seems to borrow the '7' from the horizontal line's y-value but doesn't correspond to any actual intersection point visible on the graph.

✗ **D** — $x = 8$ is close to where the parabola comes back down to the x-axis ($y = 0$), not where it crosses the horizontal line at $y = 7$ — this is reading the wrong feature of the graph.

Solving with Desmos:

Step 1: Since this is a graph-reading problem, you can use Desmos to find the actual equations first, if you want to double check. Estimate the parabola's vertex from the picture — it looks like (5, 8) — and that it passes through the origin area near (2, 0) and (8, 0). Step 2: Open Desmos and type a test parabola: y=-(x-5)^2+8

Step 3: Compare this to the picture — if it matches the shape shown, you've found the right equation.

Step 4: Type the horizontal line: y=7

Step 5: Press Enter — Desmos will show the exact intersection points when you click on them, confirming the parabola and line cross at $x = 4$ and $x = 6$, verifying choice A is a valid solution.

Nonlinear equations in one variable and systems of equations in two variables — Q4 [Hard] [Grid-In] (ID: 46308566)

Question ID: 46308566

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$r^2 + qr = 2r - 55$$

In the given equation, q is an integer constant. The given equation has no real solutions. What is the largest possible value of q?

Correct Answer: 16 (Student-Produced Response — no answer choices given)

The Rule: For a quadratic equation to have NO real solutions, its discriminant (the part under the square root in the quadratic formula, $b^2 - 4ac$) must be NEGATIVE. Rearrange any given equation into standard form ($ax^2 + bx + c = 0$) first, then set up and solve the discriminant inequality to find the range of values for an unknown constant.

The Tip: Whenever a problem tells you an equation 'has no real solutions' and asks about a range of possible values for a constant, that's your signal to use the discriminant inequality ($b^2 - 4ac < 0$) rather than trying to solve the equation directly — you can't solve it directly precisely because it has no real solutions.

Why this is the answer: Rearrange $r^2 + qr = 2r - 55$ into standard form: $r^2 + qr - 2r + 55 = 0$, which combines to $r^2 + (q-2)r + 55 = 0$. For no real solutions, the discriminant must be negative: $(q-2)^2 - 4(1)(55) < 0$, so $(q-2)^2 < 220$. Taking the square root: $-\sqrt{220} < q-2 < \sqrt{220}$. Since $\sqrt{220} \approx 14.83$, this gives approximately $-12.83 < q < 16.83$. Since q must be an integer, the largest possible value is 16.

Step-by-Step Solution:

Step 1: Start with the given equation: $r^2 + qr = 2r - 55$.

Step 2: Move all terms to one side to get standard form: $r^2 + qr - 2r + 55 = 0$.

Step 3: Combine the like terms involving r: $r^2 + (q - 2)r + 55 = 0$.

Step 4: Identify the coefficients for the discriminant formula: $a = 1$, $b = (q - 2)$, $c = 55$.

Step 5: For no real solutions, set up the discriminant inequality: $b^2 - 4ac < 0$, which becomes $(q - 2)^2 - 4(1)(55) < 0$.

Step 6: Simplify: $(q - 2)^2 - 220 < 0$, so $(q - 2)^2 < 220$.

Step 7: Take the square root of both sides (remembering this creates a two-sided inequality): $-\sqrt{220} < q - 2 < \sqrt{220}$.

Step 8: Calculate √220 ≈ 14.83.

Step 9: Add 2 to all three parts of the inequality: 2 − 14.83 < q < 2 + 14.83, which gives approximately −12.83 < q < 16.83.

Step 10: Since q is stated to be an INTEGER constant, find the largest integer within this range: the largest integer less than 16.83 is 16.

Step 11: This is a grid-in question — enter 16 as your final answer.

The Trap: A common error on this problem is forgetting to properly rearrange the equation into standard form BEFORE identifying a, b, and c — for example, using b = q instead of the correct b = (q − 2), which comes from combining the qr term with the −2r term after moving everything to one side.

Solving with Desmos:

Step 1: Open Desmos. Step 2: Type the rearranged equation with q as a slider variable: y=x^2+(q-2)x+55 (using x in place of r for graphing).

Step 3: Desmos will offer to create a slider for q — click to add it.

Step 4: Drag the slider for q and watch the parabola — when the parabola does NOT touch or cross the x-axis at all, the equation has no real solutions for that value of q.

Step 5: Slowly increase the q slider until the parabola is JUST ABOUT to touch the x-axis (this is the boundary case) — check the q value at this point; it should be very close to 16.83.

Step 6: Since q must be an integer and must stay strictly below this boundary (so the parabola never touches the x-axis), test q = 16 directly by typing y=x^2+14x+55 — confirm this parabola stays entirely above the x-axis, verifying q = 16 works and is the largest such integer.

Nonlinear equations in one variable and systems of equations in two variables — Q5 [Hard] (ID: e833de9a)

Question ID: e833de9a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$x^2 + y^2 = 36$$

$$y = mx + \frac{b}{4}$$

In the given system of equations, m and b are negative constants. In the *xy*-plane, the graphs of the equations in the given system intersect at the point $(-5, y)$, where $y < 0$. Which expression represents the value of b?

Answer

A. $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$

B. $\frac{5m}{4} - \frac{\sqrt{11}}{4}$

C. $-20m + 4\sqrt{11}$

D. $20m - 4\sqrt{11}$

Correct Answer: D

The Rule: To find where a line intersects a circle (or any curve) at a SPECIFIC given x-value, first use the circle's equation to find the corresponding y-value(s) at that x, then substitute BOTH the x and y values into the line's equation to solve for any remaining unknown constant.

The Tip: When a problem gives you a specific x-coordinate for an intersection point but leaves the y-coordinate as an unknown with a stated condition (like 'y < 0'), always solve for y using the OTHER equation in the system first (the one that doesn't contain the unknown you're solving for) — this narrows things down before you touch the equation with the unknown.

Why D is right: Substituting $x = -5$ into the circle's equation: $(-5)^2 + y^2 = 36$, which gives $25 + y^2 = 36$, so $y^2 = 11$, meaning $y = \pm\sqrt{11}$. Since the problem specifies $y < 0$, we take $y = -\sqrt{11}$. Now substitute $x = -5$ and $y = -\sqrt{11}$ into the line's equation to solve for b: $-\sqrt{11} = m(-5) + b/4$, which gives $-\sqrt{11} = -5m + b/4$. Adding 5m to both sides: $5m - \sqrt{11} = b/4$. Multiplying both sides by 4: $b = 20m - 4\sqrt{11}$.

Step-by-Step Solution:

Step 1: Substitute the known x-coordinate ($x = -5$) into the CIRCLE equation to find y: $(-5)^2 + y^2 = 36$.

Step 2: Simplify: $25 + y^2 = 36$.

Step 3: Solve for y^2: $y^2 = 36 - 25 = 11$.

Step 4: Take the square root of both sides: $y = \pm\sqrt{11}$. Since the problem states $y < 0$, choose the negative root: $y = -\sqrt{11}$.

Step 5: Now substitute both $x = -5$ and $y = -\sqrt{11}$ into the LINE equation: $y = mx + b/4$, giving $-\sqrt{11} = m(-5) + b/4$.

Step 6: Simplify the right side: $-\sqrt{11} = -5m + b/4$.

Step 7: Add 5m to both sides to isolate the b/4 term: $-\sqrt{11} + 5m = b/4$, or equivalently $5m - \sqrt{11} = b/4$.

Step 8: Multiply both sides by 4 to solve for b: $b = 4(5m - \sqrt{11}) = 20m - 4\sqrt{11}$.

Step 9: This matches choice D.

The Trap: Choice C ($-20m + 4\sqrt{11}$) is the most commonly picked wrong answer — it has the correct terms but with both signs flipped, typically resulting from a sign error while moving terms across the equals sign during Step 7, such as subtracting 5m instead of adding it.

POE — Eliminate the other three:

✗ A — This choice has both incorrect coefficients on m (should be 20, not 5) and an incorrect overall structure — it appears to divide by 4 in the wrong place.

✗ B — Similarly, this uses the wrong coefficient on m (5m instead of 20m) — likely from forgetting to multiply through by 4 in the final step.

✗ C — This has the correct magnitude for both terms but with the signs flipped ($-20m$ instead of 20m, and $+4\sqrt{11}$ instead of $-4\sqrt{11}$) — a sign error somewhere in isolating b likely while moving $-5m$ across the equation.

Solving with Desmos:

Step 1: Open Desmos and type the circle equation: x^2+y^2=36

Step 2: Press Enter — Desmos graphs a circle centered at the origin with radius 6.

Step 3: To find the y-value at $x = -5$, use the table feature on a new expression: type y=sqrt(36-x^2) and y=-sqrt(36-x^2) as two separate expressions (representing the top and bottom halves of the circle), then create a table and enter $x = -5$ for each — you'll see the negative version gives $y \approx -3.317$, matching $-\sqrt{11}$.

Step 4: Since this problem involves the unknown constants m and b, pick a simple test value for m (like $m = 1$) to make the line concrete, then type: y=1x+b/4 with a slider for b.

Step 5: Adjust the b slider until the line passes through the point $(-5, -\sqrt{11})$ (which you can also type directly as a point: (-5,-sqrt(11)) to see it plotted) — the b value where this happens should match $20(1) - 4\sqrt{11} \approx 6.73$, confirming your formula for b in terms of m matches choice D.

Nonlinear equations in one variable and systems of equations in two variables — Q6 [Hard] (ID: 5da5c665)

Question ID: 5da5c665

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$4x^2 - 5x - 5 = 0$$

What is the greatest solution to the given equation?

Answer

A. $\frac{5}{8} - \frac{\sqrt{105}}{8}$

B. $\frac{5}{8} + \frac{\sqrt{105}}{8}$

C. $\frac{5}{4} - \frac{\sqrt{105}}{4}$

D. $\frac{5}{4} + \frac{\sqrt{105}}{4}$

Correct Answer: B

The Rule: For a quadratic equation that does NOT factor into nice whole numbers, use the Quadratic Formula: $x = [-b \pm \sqrt{b^2 - 4ac}] / (2a)$. The '+' version of this formula always gives the GREATER of the two solutions, and the '−' version always gives the smaller one.

The Tip: Before reaching for the quadratic formula, quickly check whether the equation factors nicely with whole numbers (it usually doesn't when the answer choices involve square roots, like here) — square roots in the answer choices are a strong signal you should go straight to the quadratic formula rather than wasting time trying to factor.

Why B is right: Using the quadratic formula on $4x^2 - 5x - 5 = 0$ with $a = 4$, $b = -5$, $c = -5$: $x = [-(-5) \pm \sqrt{(-5)^2 - 4(4)(-5)}] / (2\times4) = [5 \pm \sqrt{25 + 80}] / 8 = [5 \pm \sqrt{105}] / 8$. The GREATEST solution uses the '+' sign: $x = 5/8 + \sqrt{105}/8$.

Step-by-Step Solution:

Step 1: Identify the coefficients from the standard form $ax^2 + bx + c = 0$: $a = 4$, $b = -5$, $c = -5$.

Step 2: Write out the quadratic formula: $x = [-b \pm \sqrt{b^2 - 4ac}] / (2a)$.

Step 3: Substitute the coefficients: $x = [-(-5) \pm \sqrt{(-5)^2 - 4(4)(-5)}] / (2(4))$.

Step 4: Simplify the numerator's first term: $-(-5) = 5$.

Step 5: Simplify inside the square root: $(-5)^2 = 25$, and $4(4)(-5) = -80$, so subtracting a negative becomes addition: $25 - (-80) = 25 + 80 = 105$.

Step 6: Simplify the denominator: $2(4) = 8$.

Step 7: The formula now reads: $x = [5 \pm \sqrt{105}] / 8$.

Step 8: Since the question asks for the GREATEST solution, use the '+' version: $x = 5/8 + \sqrt{105}/8$.

Step 9: This matches choice B.

The Trap: Choice A is the most commonly picked wrong answer — it correctly computes the formula but selects the '−' (minus) version instead of the '+' version, giving the SMALLER solution instead of the greatest one the question actually asks for.

POE — Eliminate the other three:

✗ **A** — This uses the '−' version of the quadratic formula ($5/8 - \sqrt{105}/8$), which gives the SMALLER of the two solutions, not the greatest one the question asks for.

✗ **C** — This has the correct '−' structure but with the wrong denominator (4 instead of 8 under the square root term) — a mismatched-denominator error, and also selects the smaller solution.

✗ **D** — This has the correct '+' structure but the wrong denominator under the square root term (4 instead of 8) — the square root term should be divided by 8 (matching 2a), not 4.

Solving with Desmos:

Step 1: Open Desmos and type the equation exactly as given: 4x^2-5x-5=0

Step 2: Press Enter — since this is a one-variable equation, Desmos will solve it and plot BOTH solutions as labeled points on the x-axis.

Step 3: You should see two points — one around $x \approx -0.66$ and another around $x \approx 1.9$.

Step 4: Click on the point with the LARGER x-value (the one farther to the right, around $x \approx 1.9$) — this is the greatest solution.

Step 5: To verify this matches $5/8 + \sqrt{105}/8$ exactly (not just as a decimal), type (5+sqrt(105))/8 into Desmos on a new line — it should evaluate to the same decimal value (≈1.9), confirming choice B is correct.

Nonlinear equations in one variable and systems of equations in two variables — Q7 [Hard] (ID: c8db0e19)

Question ID: c8db0e19

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$\frac{1}{4xy} + xyz = \frac{1}{3yz}$$

In the given equation, x, y, and z are positive numbers. Which expression is equivalent to y?

Answer

A. $\frac{4x-3z}{12x^2z^2}$

B. $\sqrt{\frac{4x-3z}{12x^2z^2}}$

C. $\frac{1}{3xz^2-4x^2z}$

D. $\sqrt{\frac{1}{3xz^2-4x^2z}}$

Correct Answer: B

The Rule: To solve an equation for a variable that's inside a square (like y^2, here disguised as xyz where y appears squared once isolated), first clear all fractions by multiplying through by the least common denominator, then isolate the squared variable, and finally take the square root of both sides.

The Tip: When a variable appears in a term like 'xyz' alongside fraction terms with that SAME variable in the denominator (like 1/4xy or 1/3yz), multiplying the entire equation by that variable (y, in this case) is often the fastest way to consolidate everything into y^2 terms, avoiding messy fraction-within-fraction steps.

Why B is right: Multiply every term in $1/(4xy) + xyz = 1/(3yz)$ by y: this gives $1/(4x) + xy^2z = 1/(3z)$. Subtract $1/(4x)$ from both sides: $xy^2z = 1/(3z) - 1/(4x)$. Combine the right side over a common denominator: $1/(3z) - 1/(4x) = (4x - 3z)/(12xz)$. So $xy^2z = (4x-3z)/(12xz)$. Divide both sides by xz: $y^2 = (4x-3z)/(12x^2z^2)$. Take the square root of both sides (keeping the positive root since y is positive): $y = \sqrt{[(4x-3z)/(12x^2z^2)]}$.

Step-by-Step Solution:

Step 1: Start with the given equation: $1/(4xy) + xyz = 1/(3yz)$.

Step 2: Multiply every single term in the equation by y to begin clearing y from the denominators: $y/(4xy) + xy(yz) = y/(3yz)$.

Step 3: Simplify each term: $y/(4xy) = 1/(4x)$; $xy(yz) = xy^2z$; $y/(3yz) = 1/(3z)$. The equation becomes: $1/(4x) + xy^2z = 1/(3z)$.

Step 4: Subtract $1/(4x)$ from both sides to isolate the xy^2z term: $xy^2z = 1/(3z) - 1/(4x)$.

Step 5: Combine the right side into a single fraction using the common denominator 12xz: $1/(3z) = 4x/(12xz)$, and $1/(4x) = 3z/(12xz)$, so $1/(3z) - 1/(4x) = (4x - 3z)/(12xz)$.

Step 6: Now the equation reads: $xy^2z = (4x - 3z)/(12xz)$.

Step 7: Divide both sides by xz to isolate y^2: $y^2 = (4x - 3z)/(12xz \times xz) = (4x - 3z)/(12x^2z^2)$.

Step 8: Take the square root of both sides. Since y is stated to be a positive number, keep only the positive root: $y = \sqrt{[(4x - 3z)/(12x^2z^2)]}$.

Step 9: This matches choice B.

The Trap: Choice A is the most commonly picked wrong answer — it correctly derives the expression for y^2, $(4x-3z)/(12x^2z^2)$, but forgets to take the final square root step, mistakenly presenting y^2 as if it were y itself.

POE — Eliminate the other three:

✗ **A —** This expression, $(4x-3z)/(12x^2z^2)$, is actually equal to y^2 (as derived in Step 7), NOT y itself — the crucial final square root step was skipped.

✗ **C —** This has a completely different structure (1 over a difference of terms) that doesn't match the algebra needed to isolate y from the original equation — likely from an error in combining fractions.

✗ **D —** This applies a square root to the correct 'shape' of expression (C) but that underlying expression itself is already incorrect, compounding the earlier error rather than fixing it.

Solving with Desmos:

Step 1: Since this equation has three variables (x, y, z), pick simple test values for two of them to verify your algebra — let's use x = 1 and z = 1. Step 2: Open Desmos and type the original equation with these values substituted, using y as the remaining variable: 1/(4*1*y)+1*y*1=1/(3*y*1), or simplified: 1/(4y)+y=1/(3y)

Step 3: Press Enter — Desmos will solve this one-variable equation for y and plot the positive solution as a point.

Step 4: Now test your derived formula with the same values (x=1, z=1): type sqrt((4*1-3*1)/(12*1^2*1^2)) — this should evaluate to sqrt(1/12), and you can compare this decimal value to the y-value Desmos found by solving the original equation directly — if they match, your formula (and choice B) is confirmed correct.

Nonlinear equations in one variable and systems of equations in two variables — Q8 [Hard] (ID: 9a1f1941)

Question ID: 9a1f1941

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$|4x - 3| = -9$$

How many solutions does the given equation have?

Answer

A. Zero

B. One

C. Two

D. Infinitely many

Correct Answer: A

The Rule: The absolute value of any real number is always greater than or equal to zero — an absolute value expression can NEVER equal a negative number. If an equation sets an absolute value expression equal to a negative number, that equation has NO solutions, no matter what expression is inside the absolute value bars.

The Tip: Before doing any algebra on an absolute value equation, always glance at what's on the OTHER side of the equals sign first. If that side is a negative number, you can immediately conclude there are zero solutions without any further work — this is one of the fastest 'free points' available on this type of question.

Why A is right: The equation $|4x - 3| = -9$ sets an absolute value expression equal to -9, a negative number. Since absolute value always produces a result that is zero or positive, it can never equal a negative number like -9 — so this equation has no solutions at all.

Step-by-Step Solution:

Step 1: Look at the given equation: $|4x - 3| = -9$.

Step 2: Recall the fundamental property of absolute value: |anything| is always greater than or equal to 0 — it can never be negative, because absolute value measures distance from zero, and distance is never negative.

Step 3: Since the right side of the equation is -9 (a negative number), and the left side (an absolute value expression) can never be negative, there is no possible value of x that could make this equation true.

Step 4: Conclude that the equation has zero solutions, matching choice A.

The Trap: Choice B (One) is the most commonly picked wrong answer — students sometimes rush into solving the equation algebraically (splitting into $4x - 3 = -9$ and $4x - 3 = 9$, or similar), without first checking whether the equation is even solvable in the first place, and may mistakenly think one of these branches gives a valid answer.

POE — Eliminate the other three:

✗ B — Selecting 'One' suggests attempting to solve algebraically without first recognizing that an absolute value can never equal a negative number — this check should happen BEFORE any algebra, avoiding wasted work entirely.

✗ C — Selecting 'Two' would apply the standard 'absolute value equations have two solutions' rule, but that rule only applies when the other side of the equation is POSITIVE — with a negative number on the other side, this rule doesn't apply at all.

✗ D — 'Infinitely many' would apply if the equation were something like $|4x-3| = |4x-3|$ (always true) — but this equation asks for a SPECIFIC negative value, which absolute value expressions can never produce, regardless of x.

Solving with Desmos:

Step 1: Open Desmos and type the left side of the equation as a function: y=abs(4x-3)

Step 2: Press Enter — Desmos graphs this V-shaped absolute value function, which you'll notice never dips below the x-axis (y is always ≥ 0).

Step 3: Type a second line: y=-9 — this creates a flat horizontal line far below the x-axis.

Step 4: Observe that the V-shaped graph and the horizontal line never touch or cross anywhere on the graph — no matter how far you scroll left or right, the absolute value graph stays at or above $y = 0$, while the line sits at $y = -9$.

Step 5: This visual confirms there are zero intersection points, meaning zero solutions to the equation, matching choice A.

Nonlinear equations in one variable and systems of equations in two variables — Q9 [Hard] [Grid-In] (ID: 30596346)

Question ID: 30596346

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$\frac{1}{7}(x-k)\left(16-x^2\right)=0$$

In the given equation, k is a positive constant. The sum of the solutions to the equation is 59. What is the value of k?

Correct Answer: 59 (Student-Produced Response — no answer choices given)

The Rule: For an equation already written as a product of factors set equal to zero, first find ALL the individual solutions using the Zero Product Property, then use the given information (like the sum of all solutions) to solve for any remaining unknown constant.

The Tip: When one factor is a difference of squares pattern (like $16 - x^2$), remember it factors further into TWO solutions ($x = 4$ and $x = -4$), not just one — don't forget this hidden second solution when counting up all the values that make the equation true.

Why this is the answer: Setting each factor to zero: $(x - k) = 0$ gives $x = k$; $(16 - x^2) = 0$ means $x^2 = 16$, giving $x = 4$ or $x = -4$. So the three solutions are k, 4, and −4. Since the sum of all solutions is 59: $k + 4 + (-4) = 59$, which simplifies to $k + 0 = 59$, so $k = 59$.

Step-by-Step Solution:

Step 1: Recognize the equation is already factored: $(1/7)(x - k)(16 - x^2) = 0$.

Step 2: Since 1/7 is just a nonzero constant multiplying everything, it doesn't affect where the equation equals zero — focus on the two remaining factors: $(x - k)$ and $(16 - x^2)$.

Step 3: Set the first factor equal to zero: $x - k = 0$, giving $x = k$.

Step 4: Set the second factor equal to zero: $16 - x^2 = 0$, which means $x^2 = 16$.

Step 5: Take the square root of both sides: $x = 4$ or $x = -4$ (remembering that squaring EITHER a positive or negative 4 gives 16, so BOTH are valid solutions).

Step 6: The complete solution set is $\{k, 4, -4\}$.

Step 7: Add up all the solutions and set this equal to the given sum: $k + 4 + (-4) = 59$.

Step 8: Simplify: $k + 0 = 59$, so $k = 59$.

Step 9: This is a grid-in question — enter 59 as your final answer.

The Trap: A common error on this problem is forgetting that $16 - x^2 = 0$ produces TWO solutions (both $x = 4$ and $x = -4$), not just one — a student who only finds $x = 4$ (forgetting the negative root) would set up the wrong sum equation ($k + 4 = 59$, giving an incorrect $k = 55$).

Solving with Desmos:

Step 1: Open Desmos and type the equation with k as a slider: y=(1/7)(x-k)(16-x^2)

Step 2: Desmos will offer to create a slider for k — click to add it.

Step 3: With any slider value, click on each place where the curve crosses the x-axis — you should always see THREE crossing points: two fixed ones at $x = 4$ and $x = -4$ (these don't move as you change k), and one that moves with the slider (at $x = k$).

Step 4: Adjust the k slider and watch the sum of the three x-axis crossings — you're looking for the slider value where the crossings sum to 59. Since the two fixed crossings (4 and −4) always cancel out to 0, the slider position k must itself equal 59.

Step 5: Set the slider to $k = 59$ and confirm this satisfies the problem — enter 59 into the grid-in box.

Equivalent expressions

7 questions · Easy 1 / Medium 2 / Hard 4

Equivalent expressions — Q1 [Easy] (ID: 91aa9aa9)

Question ID: 91aa9aa9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $(9x^2 + 6) - (8x^2)$?

Answer

A. 7

B. $17x^2 + 6$

C. $x^2 - 6$

D. $x^2 + 6$

Correct Answer: D

The Rule: To combine or simplify polynomial expressions, first remove the parentheses (distributing any negative sign across every term inside), then combine LIKE TERMS — terms with the exact same variable raised to the exact same power.

The Tip: When subtracting a parenthetical expression like $-(8x^2)$, remember the negative sign applies to EVERY term inside those parentheses. Even with just one term inside, it's a good habit to explicitly distribute the negative sign as a separate step to avoid sign errors on more complex problems.

Why D is right: Distribute the subtraction: $(9x^2 + 6) - (8x^2) = 9x^2 + 6 - 8x^2$. Combine the like terms (the x^2 terms): $9x^2 - 8x^2 = 1x^2 = x^2$. The equation simplifies to $x^2 + 6$.

Step-by-Step Solution:

Step 1: Write out the given expression: $(9x^2 + 6) - (8x^2)$.

Step 2: Distribute the subtraction sign across the second parentheses: $9x^2 + 6 - 8x^2$.

Step 3: Identify the like terms: $9x^2$ and $-8x^2$ are like terms (both have x^2), while 6 is a constant with no matching term to combine.

Step 4: Combine the like terms: $9x^2 - 8x^2 = x^2$.

Step 5: Write the final simplified expression: $x^2 + 6$.

Step 6: This matches choice D.

The Trap: Choice C ($x^2 - 6$) is the most commonly picked wrong answer — it correctly combines the x^2 terms but incorrectly changes the sign of the constant term, subtracting 6 instead of keeping it positive (the original +6 was never being subtracted — only the $8x^2$ term was in the second set of parentheses).

POE — Eliminate the other three:

✗ **A** — 7 completely drops the x^2 term, treating the expression as if it were only constants — this ignores that x^2 terms don't cancel out ($9x^2 - 8x^2 = x^2$, not 0).

✗ **B** — $17x^2 + 6$ adds the x^2 coefficients ($9 + 8 = 17$) instead of subtracting them — this incorrectly treats the operation between the two parentheses as addition rather than subtraction.

✗ **C** — $x^2 - 6$ correctly combines the x^2 terms but incorrectly applies the subtraction to the constant 6 as well — but the 6 was inside the FIRST parentheses (being added), not affected by the subtraction of the second parentheses.

Solving with Desmos:

Step 1: Open Desmos and type the original expression as a function: y=(9x^2+6)-(8x^2)

Step 2: Press Enter — Desmos graphs this as a parabola.

Step 3: Type your simplified answer as a second function: y=x^2+6

Step 4: Press Enter — Desmos graphs this as well. If your simplification is correct, this second curve should land EXACTLY on top of the first one (they'll appear as a single curve, since they're mathematically identical).

Step 5: As an extra check, use the table feature to compare both functions at a specific x-value, like x = 2 — both should give the same output (10), confirming the expressions are truly equivalent and verifying choice D.

Equivalent expressions — Q2 [Medium] (ID: 04887c7b)

Question ID: 04887c7b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

$$f(x) = 2x + 3$$

$$g(x) = 7x - 2$$

$$h(x) = 5x + 6$$

The functions f, g, and h are defined as shown. If $f(x) \cdot g(x) - h(x) = ax^2 + bx + c$, where a, b, and c are constants, what is the value of b?

Answer

A. -5

B. 12

C. 20

D. 22

Correct Answer: B

The Rule: To find a specific coefficient in a combined polynomial expression (like the 'b' in $ax^2 + bx + c$), you must fully multiply out (FOIL) any products first, then combine ALL like terms — including terms from a subtraction — before identifying which coefficient the question is asking about.

The Tip: When a problem gives you the FINAL form ($ax^2 + bx + c$) and asks for just one specific letter's value, do the ENTIRE multiplication and simplification first, writing out the complete simplified expression — then simply read off the coefficient the question asks for. Don't try to shortcut by guessing which original term becomes 'b.'

Why B is right: First multiply f(x) and g(x): $(2x+3)(7x-2) = 2x(7x) + 2x(-2) + 3(7x) + 3(-2) = 14x^2 - 4x + 21x - 6 = 14x^2 + 17x - 6$. Now subtract $h(x) = 5x + 6$: $(14x^2 + 17x - 6) - (5x + 6) = 14x^2 + 17x - 6 - 5x - 6 = 14x^2 + 12x - 12$. Comparing to $ax^2 + bx + c$, the coefficient of x is $b = 12$.

Step-by-Step Solution:

Step 1: Write out $f(x) \cdot g(x)$ as a multiplication: $(2x + 3)(7x - 2)$.

Step 2: Use FOIL (First, Outer, Inner, Last) to expand this product: First: $2x \times 7x = 14x^2$. Outer: $2x \times (-2) = -4x$. Inner: $3 \times 7x = 21x$. Last: $3 \times (-2) = -6$.

Step 3: Combine these four terms: $14x^2 - 4x + 21x - 6$.

Step 4: Combine the like x-terms: $-4x + 21x = 17x$. So $f(x) \cdot g(x) = 14x^2 + 17x - 6$.

Step 5: Now subtract $h(x) = 5x + 6$ from this result: $(14x^2 + 17x - 6) - (5x + 6)$.

Step 6: Distribute the negative sign across h(x): $14x^2 + 17x - 6 - 5x - 6$.

Step 7: Combine like terms: the x-terms: $17x - 5x = 12x$. The constants: $-6 - 6 = -12$.

Step 8: The fully simplified expression is: $14x^2 + 12x - 12$.

Step 9: Comparing this to the general form $ax^2 + bx + c$, we can see $b = 12$ (the coefficient of the x term).

Step 10: This matches choice B.

The Trap: Choice D (22) is the most commonly picked wrong answer — it typically results from adding the −4x and 21x terms correctly to get 17x, but then making a sign error while subtracting the 5x from h(x), perhaps adding instead of subtracting (17 + 5 = 22).

POE — Eliminate the other three:

✗ A — −5 doesn't correspond to a correct simplification path — likely from a significant sign or arithmetic error somewhere in the FOIL expansion or final combination.

✗ C — 20 could result from correctly finding 17x from the FOIL step but then making an error while combining with h(x)'s x-term, perhaps adding 5 instead of subtracting it (17 + 5 = 22, close to this but not quite matching this specific wrong value — regardless, it doesn't match the correct combination).

✗ D — 22 typically comes from adding 17 + 5 = 22 instead of correctly subtracting 17 − 5 = 12, reversing the sign of the operation when combining with h(x)'s x-coefficient.

Solving with Desmos:

Step 1: Open Desmos and type each function separately: f(x)=2x+3 then g(x)=7x-2 then h(x)=5x+6

Step 2: On a new line, type the full combined expression: y=f(x)*g(x)-h(x)

Step 3: Press Enter — Desmos graphs this as a parabola (since it now includes an x^2 term from the multiplication).

Step 4: Click the settings/wrench icon or use the table feature to inspect this parabola's equation in standard form, or simply evaluate the expression at a few x-values and compare to $14x^2+12x-12$ to confirm they match.

Step 5: As a direct check, type 14x^2+12x-12 as a separate function on another line — this should produce a curve that lands EXACTLY on top of the one from Step 2, confirming b = 12 is correct and matches choice B.

Equivalent expressions — Q3 [Medium] (ID: f8871ccd)

Question ID: f8871ccd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which expression is equivalent to $2\left(x-\frac{9}{2}\right)(x+8)$?

Answer

A. $2x^2-72$

B. $2x^2+7x-72$

C. $2x^2-x-36$

D. $x^2+7x-36$

Correct Answer: B

The Rule: To multiply an expression by a fraction times two binomials, like 2(x − 9/2)(x + 8), it's often cleanest to first combine the leading 2 with the fractional term to clear the fraction, THEN use FOIL to expand the two binomials — this avoids carrying an awkward fraction through the entire expansion.

The Tip: When you see a fraction like 9/2 paired with a leading coefficient like 2 sitting right outside the parentheses, check if multiplying them together first eliminates the fraction entirely — here, 2 × (9/2) = 9, a whole number, which makes the rest of the expansion much cleaner.

Why B is right: Distribute the 2 into the first binomial only: 2(x − 9/2) = 2x − 9. Now the expression is (2x − 9)(x + 8). Use FOIL: First: 2x × x = $2x^2$. Outer: 2x × 8 = 16x. Inner: −9 × x = −9x. Last: −9 × 8 = −72. Combine: $2x^2$ + 16x − 9x − 72 = $2x^2$ + 7x − 72.

Step-by-Step Solution:

Step 1: Start with the given expression: 2(x − 9/2)(x + 8).

Step 2: Distribute the leading 2 into the first binomial to clear the fraction: $2 \times x = 2x$, and $2 \times (-9/2) = -9$. This gives $(2x - 9)$ for the first factor.

Step 3: Rewrite the expression with this simplified first factor: $(2x - 9)(x + 8)$.

Step 4: Use FOIL to expand: First: $2x \times x = 2x^2$. Outer: $2x \times 8 = 16x$. Inner: $-9 \times x = -9x$. Last: $-9 \times 8 = -72$.

Step 5: Combine these four terms: $2x^2 + 16x - 9x - 72$.

Step 6: Combine the like x-terms: $16x - 9x = 7x$.

Step 7: The fully expanded expression is: $2x^2 + 7x - 72$.

Step 8: This matches choice B.

The Trap: Choice A ($2x^2 - 72$) is the most commonly picked wrong answer — it correctly computes the 'First' and 'Last' terms of FOIL but completely forgets to include the 'Outer' and 'Inner' cross terms ($16x$ and $-9x$), which combine to give the missing $7x$ term.

POE — Eliminate the other three:

✗ A — This is missing the middle x-term entirely ($7x$) — it looks like only the 'First' and 'Last' parts of FOIL were computed, skipping the 'Outer' and 'Inner' cross-multiplication steps.

✗ C — This has the right general shape but an incorrect middle term ($-x$ instead of $+7x$) and an incorrect constant (-36 instead of -72) — likely from distributing the leading 2 unevenly or incompletely.

✗ D — This drops the leading coefficient entirely (x^2 instead of $2x^2$) and has an incorrect constant (-36 instead of -72) — suggesting the initial distribution of the 2 was mishandled or skipped for some terms.

Solving with Desmos:

Step 1: Open Desmos and type the original expression as a function: y=2(x-9/2)(x+8)

Step 2: Press Enter — Desmos graphs this parabola.

Step 3: Type your expanded answer as a second function: y=2x^2+7x-72

Step 4: Press Enter — if your expansion is correct, this curve will land EXACTLY on top of the first one.

Step 5: As an extra check, use the table feature to compare both functions at $x = 0$ — both should output -72, and at $x = 1$, both should output $2+7-72 = -63$, confirming the expressions are equivalent and verifying choice B.

Equivalent expressions — Q4 **[Hard]** **[Grid-In]** (ID: 38c90632)

Question ID: 38c90632

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The expression $\frac{29x+102}{x(x+51)}$ is equivalent to $\frac{p}{x}+\frac{w}{x+51}$, where p and w are constants. What is the value of w?

Correct Answer: 27 (Student-Produced Response — no answer choices given)

The Rule: To split a single fraction into a sum of two simpler fractions with different denominators (called 'partial fraction decomposition'), combine the two simpler fractions back into one over their common denominator, then match the resulting numerator to the original numerator term by term.

The Tip: After combining the right side into a single fraction, line up your combined numerator directly underneath the original numerator, matching the coefficient of x with the coefficient of x, and the constant term with the constant term — this creates a small system of equations you can solve directly.

Why this is the answer: Combine p/x + w/(x+51) into a single fraction using x(x+51) as the common denominator: [p(x+51) + wx] / [x(x+51)]. Expand the numerator: px + 51p + wx = (p+w)x + 51p. Set this equal to the original numerator: (p+w)x + 51p = 29x + 102. Matching coefficients: p + w = 29 (coefficient of x) and 51p = 102 (constant term). Solving the second equation: p = 102/51 = 2. Substituting into the first equation: 2 + w = 29, so w = 27.

Step-by-Step Solution:

Step 1: Write the right side of the given equivalence, p/x + w/(x+51), as a single combined fraction using the common denominator x(x+51): [p(x+51) + w(x)] / [x(x+51)].

Step 2: Expand the numerator: p(x+51) = px + 51p, so the full numerator is px + 51p + wx.

Step 3: Combine the like terms in the numerator (the x-terms): px + wx = (p+w)x. The numerator becomes (p+w)x + 51p.

Step 4: Since this combined fraction must equal the original expression (29x+102)/[x(x+51)], and the denominators already match, set the numerators equal to each other: (p+w)x + 51p = 29x + 102.

Step 5: Match the coefficients of x on both sides: p + w = 29.

Step 6: Match the constant terms on both sides: 51p = 102.

Step 7: Solve the constant-term equation for p: p = 102/51 = 2.

Step 8: Substitute p = 2 into the x-coefficient equation: 2 + w = 29.

Step 9: Solve for w: w = 29 − 2 = 27.

Step 10: This is a grid-in question — enter 27 as your final answer.

The Trap: A common error on this problem is matching the coefficients incorrectly — for example, accidentally setting 51p equal to 29 (the x-coefficient) instead of 102 (the constant term), which comes from confusing which part of the combined numerator corresponds to the x-term versus the constant term.

Solving with Desmos:

Step 1: Open Desmos and type the original expression: y=(29x+102)/(x*(x+51))

Step 2: Press Enter — Desmos graphs this rational function.

Step 3: Type your derived answer using p = 2 and w = 27: y=2/x+27/(x+51)

Step 4: Press Enter — if your values for p and w are correct, this second curve will land EXACTLY on top of the first one.

Step 5: As an extra check, use the table feature to compare both functions at a specific x-value, like x = 1 — both should give the same output, confirming w = 27 is correct.

Equivalent expressions — Q5 [Hard] [Grid-In] (ID: 00165291)

Question ID: 00165291

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The expression $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + c$?

Correct Answer: 31/2 or 15.5 (Student-Produced Response — no answer choices given)

The Rule: When a quadratic-type expression (like $6x^4 + 31x^2 + 35$, which behaves like a quadratic in x^2) needs to be factored TWO different ways, first find its unique factorization by treating x^2 as a single variable and factoring normally, then figure out the second factorization by redistributing the SAME two roots between the two binomial factors in a different way.

The Tip: A quadratic-type expression like this has exactly two 'roots' when treated as a quadratic in x^2 — but there are multiple ways to distribute those two root-values between two binomial factors ($3x^2$ + ___)($2x^2$ + ___), depending on which root goes with which leading coefficient. This is why the problem can have BOTH an integer factorization AND a non-integer one from the very same expression.

Why this is the answer: Treating the expression as a quadratic in $y = x^2$, we have $6y^2 + 31y + 35$, which has roots $y = -5/3$ and $y = -7/2$ (found using the quadratic formula or by testing that $6(-5/3)^2 + 31(-5/3) + 35 = 0$). The integer factorization $(3x^2 + a)(2x^2 + b)$ requires matching $a/3 = 5/3$ (giving $a = 5$) and $b/2 = 7/2$ (giving $b = 7$) — both integers. The non-integer factorization $(3x^2 + c)(2x^2 + d)$ instead matches $c/3 = 7/2$ (giving $c = 21/2$) and $d/2 = 5/3$ (giving $d = 10/3$) — both non-integers, as required. So $a = 5$ and $c = 21/2$, giving $a + c = 5 + 21/2 = 10/2 + 21/2 = 31/2$.

Step-by-Step Solution:

Step 1: Notice that $6x^4 + 31x^2 + 35$ is a 'quadratic in disguise' — if you let $y = x^2$, the expression becomes $6y^2 + 31y + 35$, a standard quadratic.

Step 2: Find the roots of $6y^2 + 31y + 35 = 0$ using the quadratic formula: $y = [-31 \pm \sqrt{(31^2 - 4(6)(35))}] / (2\times6) = [-31 \pm \sqrt{(961-840)}] / 12 = [-31 \pm \sqrt{121}] / 12 = [-31 \pm 11] / 12$.

Step 3: Calculate both roots: $y = (-31+11)/12 = -20/12 = -5/3$, and $y = (-31-11)/12 = -42/12 = -7/2$.

Step 4: This means $6y^2 + 31y + 35 = 6(y + 5/3)(y + 7/2)$, which can be rewritten (redistributing the 6) as $(3y + 5)(2y + 7)$ — substituting back $y = x^2$ gives the INTEGER factorization $(3x^2 + 5)(2x^2 + 7)$, so $a = 5$ and $b = 7$.

Step 5: For the SECOND (non-integer) factorization, redistribute the SAME two roots differently between the two binomials: instead of matching 5/3 with the '3' coefficient, match 7/2 with the '3' coefficient instead: $(3x^2 + 21/2)(2x^2 + 10/3)$ — here, $c/3 = 7/2$ gives $c = 21/2$, and $d/2 = 5/3$ gives $d = 10/3$, both non-integers as required.

Step 6: Now calculate the requested value: $a + c = 5 + 21/2$.

Step 7: Convert 5 to a fraction with denominator 2: $5 = 10/2$. So $a + c = 10/2 + 21/2 = 31/2$.

Step 8: This is a grid-in question — enter 31/2 or its decimal equivalent, 15.5, as your final answer.

The Trap: A common error on this problem is only finding the integer factorization ($a = 5$, $b = 7$) and stopping there, without realizing the SAME expression has a completely different second factorization using the non-integer constants c and d — this requires recognizing that the two roots of the underlying quadratic can be distributed between the binomial factors in more than one way.

Solving with Desmos:

Step 1: Open Desmos and type the original expression: y=6x^4+31x^2+35

Step 2: Press Enter — Desmos graphs this quartic (degree-4) curve.

Step 3: Type the integer factorization to verify it: y=(3x^2+5)(2x^2+7)

Step 4: Press Enter — this curve should land exactly on top of the first one, confirming $a = 5$ and $b = 7$.

Step 5: Type the non-integer factorization using fractions: y=(3x^2+21/2)(2x^2+10/3)

Step 6: Press Enter — this curve should ALSO land exactly on top of the original, confirming $c = 21/2$ and $d = 10/3$ are correct.

Step 7: Finally, type 5+21/2 directly into Desmos — it will evaluate to 15.5, confirming a + c = 31/2 and matching your grid-in answer.

Equivalent expressions — Q6 [Hard] (ID: efc469c4)

Question ID: efc469c4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$-14(5x-2)^2+6(5x-3)^2$$

The given expression can be rewritten as $\frac{a}{6}x^2+\frac{b}{6}x+\frac{c}{6}$, where a, b, and c are constants. What is the value of $a+b+c$?

Answer

A. -612

B. -306

C. -102

D. -17

Correct Answer: A

The Rule: To expand and combine an expression with multiple squared binomials (like $-14(5x-2)^2 + 6(5x-3)^2$), first expand EACH squared binomial separately using the pattern $(p-q)^2 = p^2 - 2pq + q^2$, then distribute the leading coefficients, and finally combine all like terms together.

The Tip: Expand each squared term completely as its own separate step before multiplying by the leading coefficient (−14 or 6) — trying to combine the squaring and the coefficient distribution in one step is where most sign errors happen on problems like this.

Why A is right: Expand $(5x-2)^2 = 25x^2 - 20x + 4$, and $(5x-3)^2 = 25x^2 - 30x + 9$. Distribute: $-14(25x^2-20x+4) = -350x^2+280x-56$, and $6(25x^2-30x+9) = 150x^2-180x+54$. Add these together: $(-350+150)x^2 + (280-180)x + (-56+54) = -200x^2 + 100x - 2$. Setting this equal to $(a/6)x^2 + (b/6)x + c/6$: $a/6 = -200$ gives $a = -1200$; $b/6 = 100$ gives $b = 600$; $c/6 = -2$ gives $c = -12$. Sum: $a+b+c = -1200+600-12 = -612$.

Step-by-Step Solution:

Step 1: Expand the first squared binomial: $(5x-2)^2 = (5x)^2 - 2(5x)(2) + 2^2 = 25x^2 - 20x + 4$.

Step 2: Expand the second squared binomial: $(5x-3)^2 = (5x)^2 - 2(5x)(3) + 3^2 = 25x^2 - 30x + 9$.

Step 3: Distribute the leading coefficient −14 across the first expansion: $-14(25x^2 - 20x + 4) = -350x^2 + 280x - 56$.

Step 4: Distribute the leading coefficient 6 across the second expansion: $6(25x^2 - 30x + 9) = 150x^2 - 180x + 54$.

Step 5: Add these two results together, combining like terms: x^2 terms: $-350x^2 + 150x^2 = -200x^2$. x terms: $280x - 180x = 100x$. Constants: $-56 + 54 = -2$.

Step 6: The combined expression is: $-200x^2 + 100x - 2$.

Step 7: Set this equal to the target form $(a/6)x^2 + (b/6)x + c/6$, meaning $a/6 = -200$, $b/6 = 100$, and $c/6 = -2$.

Step 8: Solve for each constant by multiplying both sides by 6: $a = -200 \times 6 = -1200$; $b = 100 \times 6 = 600$; $c = -2 \times 6 = -12$.

Step 9: Calculate the requested sum: $a + b + c = -1200 + 600 + (-12) = -612$.

Step 10: This matches choice A.

The Trap: Choice B (−306) is the most commonly picked wrong answer — it's exactly HALF of the correct answer (−612), which typically results from forgetting to multiply by 6 when solving for a, b, and c individually (i.e., using $a = -200$, $b = 100$, $c = -2$ directly as if they were already the final values, then only partially adjusting).

POE — Eliminate the other three:

✗ **B —** −306 is exactly half of the correct total — this often results from a computational shortcut gone wrong, such as only multiplying some terms by 6 instead of all three constants consistently.

✗ **C —** −102 doesn't correspond to a clean derivation from this problem — likely a distractor value or the result of multiple compounding errors.

✗ **D —** −17 is far too small in magnitude to result from correctly scaling by 6 — this likely comes from forgetting to multiply by 6 at all and instead just adding −200 + 100 + (−2) = −102 (close to choice C) or a similar unscaled combination.

Solving with Desmos:

Step 1: Open Desmos and type the original expression: y=-14(5x-2)^2+6(5x-3)^2

Step 2: Press Enter — Desmos graphs this parabola.

Step 3: Type your simplified expression using the target form with the constants you found: y=(-1200/6)x^2+(600/6)x+(-12/6)

Step 4: Press Enter — if your values for a, b, and c are correct, this curve will land EXACTLY on top of the original one.

Step 5: As a final check, type -1200+600-12 directly into Desmos — it will evaluate to −612, confirming a+b+c = −612 and matching choice A.

Equivalent expressions — Q7 [Hard] (ID: 2289b199)

Question ID: 2289b199

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$\frac{\sqrt[3]{n^{17}p^5}}{5n^2\left(\sqrt[8]{p^8}\right)}$$

For all positive values of n and p, the given expression is equivalent to which of the following?

I. $\frac{\left(n^{\frac{14}{3}}\right)\left(p^{\frac{5}{3}}\right)}{5np}$

II. $\frac{\sqrt[3]{n^{11}p^2}}{5}$

Answer

A. I only

B. II only

C. I and II

D. Neither I nor II

Correct Answer: C

The Rule: To simplify an expression involving roots (radicals) with different index numbers (like a cube root and an eighth root), first rewrite every root as a FRACTIONAL EXPONENT (the nth root of x becomes x^(1/n)), then use standard exponent rules to combine and simplify everything into one unified expression.

The Tip: Whenever you see mixed radical types (cube roots, eighth roots, etc.) in the same expression, immediately convert ALL of them to fractional exponent notation — this turns a confusing-looking radical problem into a much more familiar exponent-simplification problem you already know how to handle.

Why C is right: Rewrite the original expression using fractional exponents: the numerator ∛(n^17p^5) becomes n^(17/3)p^(5/3), and the denominator's $\sqrt[8]{}$(p^8) becomes p^(8/8) = p^1 = p. So the full expression becomes [n^(17/3)p^(5/3)] / [$5n^2p$] = (1/5) n^(17/3−2) p^(5/3−1) = (1/5) n^(11/3) p^(2/3). Checking option I: [n^(14/3)p^(5/3)]/(5np) = (1/5)n^(14/3−1)p^(5/3−1) = (1/5)n^(11/3)p^(2/3) — matches! Checking option II: ∛(n^11p^2)/5 = (1/5)n^(11/3)p^(2/3) — also matches! Both expressions are equivalent to the original.

Step-by-Step Solution:

Step 1: Rewrite the numerator using fractional exponent notation: $\sqrt[3]{n^{17}p^5} = (n^{17}p^5)^{1/3} = n^{17/3}p^{5/3}$.

Step 2: Rewrite the denominator's radical term: $\sqrt[8]{p^8} = p^{8/8} = p^1 = p$ (since 8 divided by 8 equals exactly 1).

Step 3: Write the full original expression using these simplified forms: $[n^{17/3}p^{5/3}] / [5n^2p]$.

Step 4: Simplify by subtracting exponents for matching bases (dividing powers with the same base means subtracting exponents): for n: $17/3 - 2 = 17/3 - 6/3 = 11/3$. For p: $5/3 - 1 = 5/3 - 3/3 = 2/3$.

Step 5: The fully simplified original expression is: $(1/5)\, n^{11/3}\, p^{2/3}$.

Step 6: Check option I: $[n^{14/3}p^{5/3}] / (5np)$. Simplify exponents: for n: $14/3 - 1 = 14/3 - 3/3 = 11/3$. For p: $5/3 - 1 = 5/3 - 3/3 = 2/3$. This gives $(1/5)n^{11/3}p^{2/3}$ — an EXACT match to Step 5.

Step 7: Check option II: $\sqrt[3]{n^{11}p^2}/5 = (n^{11}p^2)^{1/3}/5 = n^{11/3}p^{2/3}/5$ — also an EXACT match to Step 5.

Step 8: Since BOTH option I and option II are algebraically equivalent to the original expression, the correct answer is C: I and II.

The Trap: Choice A (I only) is the most commonly picked wrong answer — students sometimes correctly verify option I matches but assume option II (written with a cube root instead of separated exponents) must be a 'different' or 'trick' expression, without actually converting it to fractional exponent form to check.

POE — Eliminate the other three:

✗ A — While option I does correctly match the original expression, option II ALSO matches when converted to fractional exponent form — dismissing option II without checking it carefully is the error here.

✗ B — While option II does correctly match, option I ALSO matches — both are valid equivalent forms of the same original expression, so 'II only' incorrectly excludes a correct option.

✗ D — Since BOTH option I and option II simplify to the exact same expression as the original (as shown in Steps 6–7), 'Neither I nor II' is incorrect — both options are actually valid.

Solving with Desmos:

Step 1: Since this involves variables n and p in exponents, pick simple positive test values to verify — let's use $n = 8$ and $p = 4$ (chosen because they're powers of 2, which often simplify nicely with fractional exponents). Step 2: Open Desmos and type the original expression with these values: (8^(17/3)*4^(5/3))/(5*8^2*(4^(8/8)))

Step 3: Press Enter — Desmos will evaluate this to a specific decimal number.

Step 4: Type option I with the same values: (8^(14/3)*4^(5/3))/(5*8*4)

Step 5: Press Enter — compare this result to Step 3's result; they should be IDENTICAL, confirming option I matches.

Step 6: Type option II with the same values: (8^11*4^2)^(1/3)/5

Step 7: Press Enter — compare this to Step 3's result as well; it should ALSO be identical, confirming option II matches too, verifying choice C (I and II) is correct.